

Project Interim Report

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Objectives of the Study

The main objective of this project is to design and develop an AI-powered Autonomous Cyber Defense System for real-time threat hunting in Cloud ERP environments.

Specific objectives include:

- To identify and analyze cyber threats in cloud-based ERP systems
 - To develop an AI-based model for detecting anomalies and malicious activities
 - To automate threat detection and response mechanisms
 - To enhance real-time monitoring and proactive defense capabilities
 - To improve overall security posture of cloud ERP systems
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Scope of the Study

The scope of this study is focused on:

- Cloud-based ERP environments and their security challenges
- Real-time cyber threat detection using Artificial Intelligence
- Development of a prototype system for automated threat hunting
- Analysis of attack patterns such as malware, intrusion, and unauthorized access

Limitations:

- The project is limited to a prototype-level implementation
 - Real-world deployment and large-scale testing are not included
 - Data availability for training AI models may be limited
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Methodology

This research follows an exploratory and applied research approach.

The methodology includes:

1. Understanding cyber threats in cloud ERP systems
2. Collecting dataset (synthetic or publicly available cybersecurity datasets)
3. Designing AI models for anomaly detection
4. Developing a prototype system

5. Testing and validating the system

Research Design

The research design is exploratory and experimental in nature.

- Exploratory: To understand and analyze cyber threats
- Experimental: To build and test AI-based detection models

The study focuses on developing a working prototype and evaluating its effectiveness in threat detection.

Data Collection Method

The data used in this study will be:

- Secondary Data:
 - Public cybersecurity datasets
 - Research papers and journals
 - Online threat intelligence sources
 - Synthetic Data:
 - Simulated attack scenarios for testing the system
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Sampling Method (if applicable)

- Random sampling of cybersecurity datasets will be used
 - Selection of relevant data related to:
 - Network traffic
 - System logs
 - User activity
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Data Analysis Tools

The following tools and technologies will be used:

- Programming Languages: Python
- Libraries/Frameworks: TensorFlow, Scikit-learn, Pandas, NumPy

- Cloud Platform: AWS / Azure (for simulation)
- Security Tools: SIEM tools (basic level), log analyzers
- Visualization Tools: Matplotlib / Power BI